When to use which mean?

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Some slides are courtesy of other professors

Arithmetic mean

$$\overline{x}_A = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Harmonic mean

$$\overline{x_H} = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}}$$

Geometric mean

$$\overline{x_G} = \sqrt[n]{x_1 x_2 \cdots x_i \cdots x_n}$$

$$= \left(\prod_{i=1}^{n} x_i \right)^{1/n}$$

Weighted means

$$\sum_{i=1}^{n} w_i = 1$$

$$\bar{x}_A = \sum_{i=1}^{n} w_i x_i$$

$$\bar{x}_H = \frac{1}{\sum_{i=1}^{n} \frac{w_i}{x_i}}$$

- Standard definition of mean assumes all measurements are equally important
- Instead, choose weights to represent relative importance of measurement i

Which is the right mean?

- Arithmetic (AM)?
- Harmonic (HM)?
- Geometric (GM)?
- WAM, WHM?
- Which one should be used when?

What makes a good mean?

- *Time*–based mean (e.g. seconds)
  - Should be *directly proportional* to total time (sum of time of sample values)
  - If time value of every sample doubles, mean value should also double
- *Rate*–based mean (e.g. operations/sec)
  - Should be *inversely proportional* to total time (sum of time of sample values)
  - If time value of every sample doubles, mean value should reduce by half
- Which means satisfy these criteria?

Solved question: Finding the right mean

- We drive 30 mph for first 10 miles, then 90 mph for next 10 miles, what is average speed?
- Average speed = (30+90)/2 **WRONG**
- Average speed = total distance / total time

  = (20 / (10/30 + 10/90))

  = 45 mph
- When dealing with *rates* (mph) do not use arithmetic mean

Arithmetic mean for times

- Produces a mean value that is *directly proportional to total time*

→ Correct mean to summarize *execution time*

$$\overline{T_A} = \frac{1}{n}\sum_{i=1}^{n} T_i$$

Arithmetic mean for rates

- Produces a mean value that is proportional to *sum of inverse of times*
- But we want *inversely proportional to sum of times*

$$\overline{M_A} = \frac{1}{n}\sum_{i=1}^{n} M_i$$

$$= \sum_{i=1}^{n} \frac{F / T_i}{n}$$

$$= \frac{F}{n}\sum_{i=1}^{n} \frac{1}{T_i}$$

Arithmetic mean for rates

- Produces a mean value that is proportional to *sum of inverse of times*
- But we want *inversely proportional to sum of times*

→ Arithmetic mean is ***NOT*** appropriate for summarizing rates

$$\overline{M}_A = \frac{1}{n}\sum_{i=1}^{n} M_i$$

$$= \sum_{i=1}^{n} \frac{F/T_i}{n}$$

$$= \frac{F}{n}\sum_{i=1}^{n} \frac{1}{T_i}$$

Harmonic mean for times

- Not directly proportional to *sum of times*

$$\overline{T_H} = \frac{n}{\sum_{i=1}^{n} \frac{1}{T_i}}$$

Harmonic mean for times

- Not directly proportional to *sum of times*

→ Harmonic mean is ***not*** appropriate for summarizing times

$$\overline{T}_H = \frac{n}{\sum_{i=1}^{n} \frac{1}{T_i}}$$

Harmonic mean for rates

- Produces
  (total number of ops)
  ÷ (sum execution times)
- Inversely proportional to total execution time

→ Harmonic mean is appropriate to summarize rates

$$\overline{M_H} = \frac{n}{\sum_{i=1}^{n} \frac{1}{M_i}} = \frac{n}{\sum_{i=1}^{n} \frac{T_i}{F}} = \frac{Fn}{\sum_{i=1}^{n} T_i}$$

Harmonic mean for rates

Sec	10^9 FLOPs	MFLOPS
321	130	405(=130000/321)
436	160	367
284	115	405
601	252	419
482	187	388

$$\overline{M_H} = \frac{5}{\left(\frac{1}{405}+\frac{1}{367}+\frac{1}{405}+\frac{1}{419}+\frac{1}{388}\right)}$$

$$= 396$$

$$\overline{M_H} = \frac{844 \times 10^9}{2124} = 396$$

130+160+115+252+187 = 844
321+436+284+601+482 = 2124

Solved Question

Q. Three applications P, Q, R have a transactions rate of 54 per minute, 87 per minute and 99 per minute respectively. They run one after another. If each of them runs for 600 seconds, find the correct average value of transactions per minute.

Also write which mean would you use to get the average

Answer. Since each of them ran for the same amount of time, we can use arithmetic mean.

Average = (54+87+99)/3 = 80 transactions per minute

Geometric Mean

- Use it for ratios (unitless quantities)
- Geometric mean of ratios =

$$\sqrt[n]{\prod_{i=1}^{n} ratio(i)}$$

- Independent of reference machine

Geometric mean

- Correct mean for averaging normalized values
- Good when averaging measurements with wide range of values
- Maintains consistent relationships when comparing normalized values
  - Independent of basis used to normalize

Geometric mean with times

	System 1	System 2	System 3
	417	244	134
	83	70	70
	66	153	135
	39,449	33,527	66,000
	772	368	369
Product	6.95E+13	3.22E+13	3.08E+13
Geo mean	587	503	499
Rank	3	2	1

Geometric mean normalized to System 1

	System 1	System 2	System 3
	1.0	0.59	0.32
	1.0	0.84	0.85
	1.0	2.32	2.05
	1.0	0.85	1.67
	1.0	0.48	0.45
Geo mean	1.0	0.86	0.84
Rank	3	2	1

Geometric mean normalized to System 2

	System 1	System 2	System 3
	1.71	1.0	0.55
	1.19	1.0	1.0
	0.43	1.0	0.88
	1.18	1.0	1.97
	2.10	1.0	1.0
Geo mean	1.17	1.0	0.99
Rank	3	2	1

Sum of execution times

	System 1	System 2	System 3
	417	244	134
	83	70	70
	66	153	135
	39,449	33,527	66,000
	772	368	369
Sum	40,787	34,362	66,798
Arith mean	8157	6872	13,342
Rank	2	1	3

What's going on here?!

	System 1	System 2	System 3
Geo mean wrt 1	1.0	0.86	0.84
Rank	3	2	1
Geo mean wrt 2	1.17	1.0	0.99
Rank	3	2	1
Arith mean	8157	6872	13,342
Rank	2	1	3

Geometric mean for times

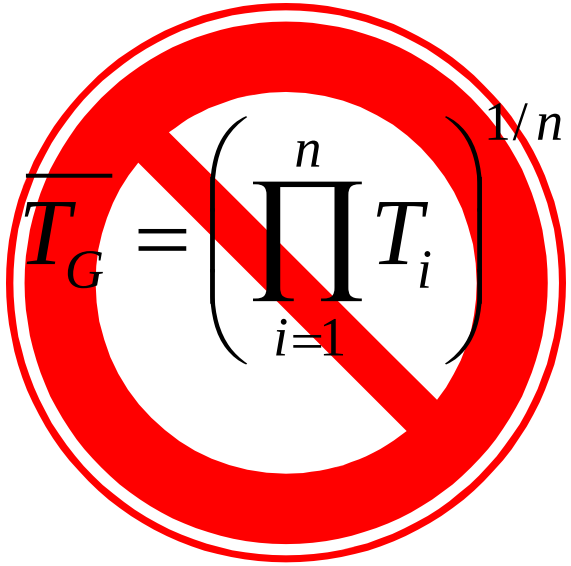
- Not directly proportional to *sum of times*

$$\overline{T_G} = \left(\prod_{i=1}^{n} T_i \right)^{1/n}$$

Geometric mean for times

- Not directly proportional to *sum of times*

→ Geometric mean is ***not*** appropriate for summarizing times



Geometric mean for rates

- Not inversely proportional to *sum of times*

$$\overline{T_G} = \left(\prod_{i=1}^{n} M_i \right)^{1/n}$$

$$= \left(\prod_{i=1}^{n} \frac{F}{T_i} \right)^{1/n}$$

Geometric mean for rates

- Not inversely proportional to *sum of times*

→ Geometric mean is ***not*** appropriate for summarizing rates

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Summary of Means

- Avoid means if possible
  - Loses information
- Arithmetic
  - When sum of raw values has physical meaning
  - Use for summarizing **times** (not rates)
- Harmonic
  - Use for summarizing **rates** (not times)
- Geometric mean
  - Not useful when *time* is best measure of perf

Geometric mean

- Does provide consistent rankings
  - Independent of basis for normalization
- But can be consistently wrong!
- Value can be computed
  - But has no physical meaning

AM? GM? HM? WAM? WHM? What are the Weights??????

Measure	Valid central tendency for summarized measure over the suite	
IPC		
CPI		
Speedup		
MIPS		
MFLOPS		
Cache hit rate		
Cache misses per instruction		
Branch misprediction rate per branch		
Normalized execution time		
Transactions per minute		
A/B		

Conditions under which unweighted arithmetic and harmonic means are valid indicators of overall performance

Measure	When is AM valid?	When is H.M. valid?
A/B	If B's are equal	If A's are equal
IPC	If equal cycles in each benchmark	If equal work (I-count) in each benchmark
CPI	If equal I-count in each benchmark	If equal cycles in each benchmark
Speedup	If equal execution times in each benchmark in the improved system	If equal execution times in each benchmark in the baseline system
MIPS	If equal times in each benchmark	If equal I-count in each benchmark
MFLOPS	If equal times in each benchmark	If equal FLOPS in each benchmark
Cache hit rate	If equal number of references to cache for each benchmark	If equal number of cache hits in each benchmark
Cache misses per instruction	If equal I-count in each benchmark	If equal number of misses in each benchmark
Normalized execution time	If equal execution times in each benchmark in the system considered as base	If equal execution times in each benchmark in the system being evaluated
Transactions per minute	If equal times in each benchmark	If equal number of transactions in each benchmark

What do we mean by “valid” in previous slide?

- Given a series of sub-trips at different speeds, if each sub-trip covers the same distance, then the average speed is the harmonic mean of all the sub-trip speeds

- If each sub-trip takes the same amount of time, then the average speed is the arithmetic mean of all the sub-trip speeds

- **Case 1: Same distance**
- Assume we drove 1m for 2s, 1m for 4s and 1m for 5s
- Total dist = 3m, total time = 11s, speed = 3/11 = 0.27
- AM → (1/2 + ¼+1/5)/3 = 0.95/3 = 0.3166 WRONG
- HM -> (3/(2/1+4/1+5/1)) = 3/11 = 0.27 RIGHT

- **Case 2: Same time**
- Assume we drove 10m for 20s, 5m for 20s and 4m for 20s
- Total dist = 19m, total time = 60s, speed = 19/60 = 0.3166
- AM → (10/20 + 5/20+4/20)/3 = 0.95/3 = 0.3166 RIGHT
- HM -> (3/(20/10+20/5+20/4)) = 3/11 = 0.27 WRONG

The mean to be used to find aggregate measure over a benchmark suite from measures corresponding to individual benchmarks in suite

Measure	Valid central tendency for summarized measure over the suite	
IPC	W.A.M. weighted with cycles	W.H.M. weighted with I-count
CPI	W.A.M. weighted with I-count	W.H.M. weighted with cycles
Speedup	W.A.M. weighted with execution time ratios in improved system	W.H.M. weighted with execution time ratios in the baseline system
MIPS	W.A.M. weighted with time	W.H.M. weighted with I-count
MFLOPS	W.A.M. weighted with time	W.H.M. weighted with FLOP count
Cache hit rate	W.A.M. weighted with number of references to cache	W.H.M. weighted with number of hits
A/B	W.A.M. weighted with B's	W.H.M. weighted with A's

Weighted means

$$\sum_{i=1}^{n} w_i = 1$$

$$\bar{x}_A = \sum_{i=1}^{n} w_i x_i$$

$$\bar{x}_H = \frac{1}{\sum_{i=1}^{n} \frac{w_i}{x_i}}$$

- Standard definition of mean assumes all measurements are equally important
- Instead, choose weights to represent relative importance of measurement i

- Assume we drove 1m for 2s, 3m for 4s and 5m for 6s
- Total dist = (1+3+5)=9m, total time = (2+4+6)=12s, speed = 9/12 = 0.75
- Unweighted AM → (1/2 + 3/4+5/6)/3 = 0.69 WRONG
- Unweighted HM -> (3/(2/1+4/3+ 6/5)) = 0.66 WRONG

- **Weighted AM.** Weights are 2/12, 4/12 and 6/12
- AM = (2/12)*(1/2)+ (4/12)*(3/4)+(6/12)*(5/6)

=> AM = 1/12 + 3/12 + 5/12 = 9/12 = 0.75 RIGHT

- Weighted HM. Weights are 1/9, 3/9 and 5/9
- HM = $\frac{1}{(\frac{1}{9}*\frac{2}{1}+\frac{3}{9}*\frac{4}{3}+\frac{5}{9}*\frac{6}{5})} = \frac{9}{2+4+6} = \frac{9}{12} = 0.75$ RIGHT

Reference

- Excellent discussion of 3 types of means: https://towardsdatascience.com/on-average-youre-using-the-wrong-average-geometric-harmonic-means-in-data-analysis-2a703e21ea0